

THREE LOCAL INVERSE POINTS, TWO RAMIFICATION TYPES, AND AN ORIGINAL ERDŐS–STRAUS CHANNEL TRANSFER

THE CLANKERS

ABSTRACT. The reciprocal cubic of an original prime Erdős–Straus state is a literal target of the four-dimensional ES–Fabel polynomial. Its seven-point affine fibre always has exactly three rational points over the original local field. Exterior states give two ramified quadratic factors; middle states give two unramified quadratic factors. The original prime character distinguishes the algebras although rational-point counts do not. An explicit negative-square shape transfer changes that arithmetic target at the same prime and returns a middle state. Universal existence is not asserted.

1. ORIGINAL COORDINATES

Fix a prime $p \equiv 1 \pmod{4}$, $p/4 < a < p/2$, $R = 4a - p$, and $u \mid a^2$. The original gates are $E : R \mid 4u + 1$ and $M : R \mid u + a$. Set $g = \gcd(a, u)$, $(h, r, s) = (g^2/u, u/g, a/g)$. For E let $\kappa = (pr + s)/R$ and $(x_1, x_2, x_3) = (a, h\kappa, phr\kappa)$; for M let $\lambda = (r + s)/R$ and $(x_1, x_2, x_3) = (a, phs\lambda, phr\lambda)$. These are the classical Type-I/II coordinates [1, §2]. They give $4/p = \sum_i 1/x_i$, and their ordered inverses are $u = a^2/(Rx_2 - pa)$ and $u = pa^2/(Rx_2 - pa)$.

Lemma 1. *Every original square divisor has $(u/R) = (u/p)$. In particular E implies $(h/p) = -1$, and M implies $(rs/p) = -1$.*

Proof. For odd $l \mid a$, reciprocity gives $(l/R) = (-1/l)(R/l) = (p/l) = (l/p)$. For $2 \mid a$, use $R \equiv -p \pmod{8}$ and the supplementary law. Multiply over the original exponents. The E target is $-1/4$; for M use $r \equiv -s \pmod{R}$ and $(-1/R) = -1$. $\square$

The ordered reciprocal roots $t_i = p/x_i$ are distinct and have sum four. Retain the cubic and its quartic target

$$f(T) = \prod_i (T - t_i) = T^3 - 4T^2 + \sigma_2 T - \sigma_3, \quad H(U, V) = V \prod_i (U - t_i V).$$

The coefficient map to the existing ES–Fabel polynomial is $\mathbf{U} = (0, -4, \sigma_2, -\sigma_3)$. Its full marked arithmetic inverse is $x_i = p/t_i$, subject to positive integrality, followed by the original divisor inverse above. Neither p nor the ordering is discarded.

2. THE ORIGINAL POLYNOMIAL FIBRE

For the exact four-dimensional polynomial P in [2, GF1], its factor coordinates have $L = \xi U + bV$ and $C = cU^3 + dU^2V + eUV^2 + f_0V^3$, with $\xi d + bc = 1$ and $C(-b, \xi) = 1$; their product is $P_0U^4 + U^3V + P_2U^2V^2 + P_3UV^3 + P_4V^4$. For $d_i = f'(t_i)$, the full finite inverse is

$$\begin{aligned} \xi_i^2 &= d_i^{-1}, \quad y_i = -t_i - i/\xi_i, \quad z_i = 2t_i/\xi_i + 3i/\xi_i^2, \\ w_i &= 7it_i^2/\xi_i + (-4 - 17t_i)/\xi_i^2 - 13i/\xi_i^3. \end{aligned} \tag{1}$$

Both signs occur. This follows by factoring at $t_i = -b/\xi_i$ and evaluating $C(-b, \xi_i) = \xi_i^2 f'(t_i)$. The factor-coordinate inverse in [2, GF5–7] gives (1). The single infinity chart is

$$(0, -4, i(112 - \sigma_2)/2, -704 + 8\sigma_2 + \sigma_3). \tag{2}$$

The opposite projective factor sign at infinity is absent from this particular affine chart. Thus there are seven geometric points. The polynomial Jacobian is identically -2 ; its original constant is retained.

Fix either embedding $i \mapsto i_p \in \mathbb{Q}_p$. The complete fibre algebra is

$$\mathcal{A} = \mathbb{Q}_p \times \prod_{i=1}^3 \mathbb{Q}_p[\xi_i]/(\xi_i^2 - d_i^{-1}). \quad (3)$$

Theorem 1. *Every original E or M state has exactly three $\mathbb{Q}_p$ -rational inverse points. Its complete algebra is*

$$E : \mathbb{Q}_p^3 \times K_{\text{ram}}^2, \quad M : \mathbb{Q}_p^3 \times K_{\text{unr}}^2,$$

with respectively a ramified and the unramified quadratic extension. The two copies retain different original root labels.

Proof. For E let $H = hrs\kappa$. The roots are $(p\kappa, pr, s)/H$, with H a p -unit. The difference $\kappa - r$ is also a unit: otherwise $pD + 1 = 4hr\kappa$ would make the nonresidue hr^2 equal to $1/4 \pmod{p}$. Consequently the derivative valuations are $(1, 1, 0)$, $d_1/d_2 \equiv -1 \pmod{p}$ and $d_3 \equiv 16 \pmod{p}$. Unit square residues lift to squares, so the first two factors are copies of one ramified quadratic field and the last splits.

For M let $H = hrs\lambda$. The roots are $(p\lambda, r, s)/H$, with distinct reductions. Up to H^{-2} , their derivatives reduce to $rs, r(r-s), s(s-r)$. The first is a nonsquare by the lemma. Their product is the square $-r^2s^2(r-s)^2$, so precisely one is a square and two are nonsquare units. The latter define the same unramified quadratic field. Add the single infinity factor in both cases. The standard square-unit lifting step is explicitly $b_{n+1} = b_n + p^n t$, with $t = -(b_n^2 - z)p^{-n}(2b_n)^{-1} \pmod{p}$. $\square$

In E, $\bar{f} = T^2(T-4)$, $v_p(\sigma_2, \sigma_3, \text{Disc } f) = (1, 2, 2)$. Formula (1) gives valuation vector $(-1/2, 1/2, 1, 1)$ for all four branches above the first two roots. They are not integral. Only the two branches at four and the infinity point specialize, leaving three geometric special points. In M all seven branches are integral over unramified extensions, and seven geometric special points remain. Both special fibres nevertheless have three $\mathbb{F}_p$ -rational points. The difference is not a zero source Jacobian: -2 is still a unit.

The original complex conductor conjugacy in [2, FC32–35] transports all seven points and their original norms. It is not thereby a $\mathbb{Q}_p$ -model of its complex period matrix. Here the arithmetic statements apply to the literal polynomial over $\mathbb{Q}(i)$ with the specified embedding.

The integral root order records the lost arithmetic. Evaluation identifies the E root order with $\{z \in \mathbb{Z}_p^3 : z_1 \equiv z_2 \pmod{p}\}$, of index p and conductor $p\mathbb{Z}_p \times p\mathbb{Z}_p \times \mathbb{Z}_p$; the M order is all $\mathbb{Z}_p^3$. Necessity follows from congruence of the first two E roots. Sufficiency follows by integral interpolation at those two roots and a correction by $(T-t_1)(T-t_2)$ at the separated third root.

For any same-prime E and M root triples x_i, y_i , the unique labelled quadratic interpolation $F(x_i) = y_i$ is $F(T) = \sum_i y_i(T-x_j)(T-x_k)/[(x_i-x_j)(x_i-x_k)]$. It has coefficient valuations $(0, -1, -1)$ and $pF \pmod{p}$ is a nonzero multiple of $T(T-4)$. The reverse interpolation is integral. The claims follow directly from the derivative valuations: after multiplying by p , only the second summand survives modulo p . Thus a rational root-level isomorphism and its precise integral obstruction both exist. This root-order conductor is distinct from the weighted complex conductor in the Zeta source.

3. CHANGING THE ARITHMETIC TARGET AT THE SAME PRIME

For E put $v = hs^2$, $D = (4u+1)/R$, $\alpha = v - R$, $\beta = v - D$. Direct reduction gives

$$h \mid \alpha\beta - 1, \quad p \equiv \alpha \pmod{h}, \quad p\beta \equiv 1 \pmod{h}.$$

Theorem 2. *If $\alpha = -4c^2$ or $\beta = -4c^2$ for a positive integer c , then $h \equiv 3 \pmod{4}$ and $\gcd(h, c) = 1$. Put $j = (h+1)/4$. On the exact domain $c \mid j$, set*

$$Q = h, \quad U = c^2 \text{ (}\alpha \text{ case)}, \quad U = (j/c)^2 \text{ (}\beta \text{ case)}, \quad R_M = (p+4U)/h, \quad a_M = (p+R_M)/4.$$

This is an original M state with

$$g_M = \gcd(j, U), \quad (h_M, r_M, s_M, \lambda_M) = (g_M^2/U, U/g_M, R_M j/g_M - U/g_M, j/g_M).$$

In particular both entire lines $\alpha = -4$ and $\beta = -4$ have middle returns.

Proof. The shape congruences make h odd and coprime to c . Reciprocity gives $(h/p) = (-1/h) = -1$, hence $h = 4j - 1$. In the first case $h \mid p + 4c^2$. In the second, $p \equiv -(4c^2)^{-1}$ and $4j \equiv 1 \pmod{h}$ give $h \mid p + 4(j/c)^2$. The stated divisibility is precisely what puts the chosen word in the actual box $U \mid j^2$. For the first case $4U < p - h$ follows from the original $R = hs^2 + 4c^2 < p$. For the second, $4U \leq (h+1)^2/4 < (h-1)p$, since $p > 2h$ and $h \geq 3$. Thus $0 < R_M < p$. The displayed gcd normalization gives $a_M = h_M r_M s_M$, $U = h_M r_M^2$ and $R_M \lambda_M = r_M + s_M$. Any prime dividing r_M, R_M would divide p , impossible because $r_M \leq j < p$. The normalization is primitive and the ordered M identity follows. With $j = cm$, $d = \gcd(c, m)$, the new grade is explicitly $h_M = d^2$. $\square$

At the hard prime $p = 5209$, the original E state $(a, R, u, h, r, s, \kappa) = (1330, 111, 18620, 95, 14, 1, 657)$ has shape $(-16, -576)$. Its two marked returns ($c = 2$ on α , $c = 12$ on β) coincide at $j = 24, U = 4$. The M state is $(a_M, R_M, u_M, h_M, r_M, s_M, \lambda_M) = (1316, 55, 4, 4, 1, 329, 6)$, giving

$$\frac{4}{5209} = \frac{1}{1316} + \frac{1}{41130264} + \frac{1}{125016}.$$

The actual new factorization contains 47, absent from the old E source. The local fibre changes from ramified to unramified because the target has changed, not through an impossible isomorphism of the unchanged local algebras.

Availability cannot be suppressed. At $p = 3049$, the E state $(806, 175, 20956, 31, 26, 1, 453)$ has $\alpha = -144$, but $c = 6$ does not divide $j = 8$. No divisor of 64 is the required 5 (mod 31), so the entire selected cofactor box is empty. Another original M shell, $a = 770, u = 5, R = 31$, succeeds.

4. PROOF BOUNDARY AND CERTIFICATE

The full source lists both inverse fibre parametrizations and all coefficient collisions. The standard-library certificate retains the original exponents, ordered denominators, all signed Fabel coordinates, Jacobian checks and Hensel lifts. A separate primitive-parameter implementation reproduces the complete bounded source and uses the original factor equations instead of the expanded polynomial. An inverse at an arbitrary cubic target does not prove an ES state: $(0, -4, 0, 1)$ has infinity inverse $(0, -4, 56i, -705)$ but its cubic has no rational root. Universal original target supply remains unresolved.

REFERENCES

- [1] Elsholtz, C., & Tao, T. (2013). Counting the number of solutions to the Erdős–Straus equation on unit fractions. *J. Aust. Math. Soc.*, 94, 50–105. [arXiv:1107.1010](#), §2.
- [2] Split-Zero programme. (2026). *The ES–Fable inverse correspondence in the original weighted conductor*. GF1–11, GF16–18, FC32–35. Zeta workbench revision [369f6ea9e0312bdf348ca9efd294735efc99c51d](#), source blob [f777bb9f1434fa90f2f8bc762860048cc1d32bb4](#). The four-dimensional polynomial and inverse are antecedents, not newly claimed here.